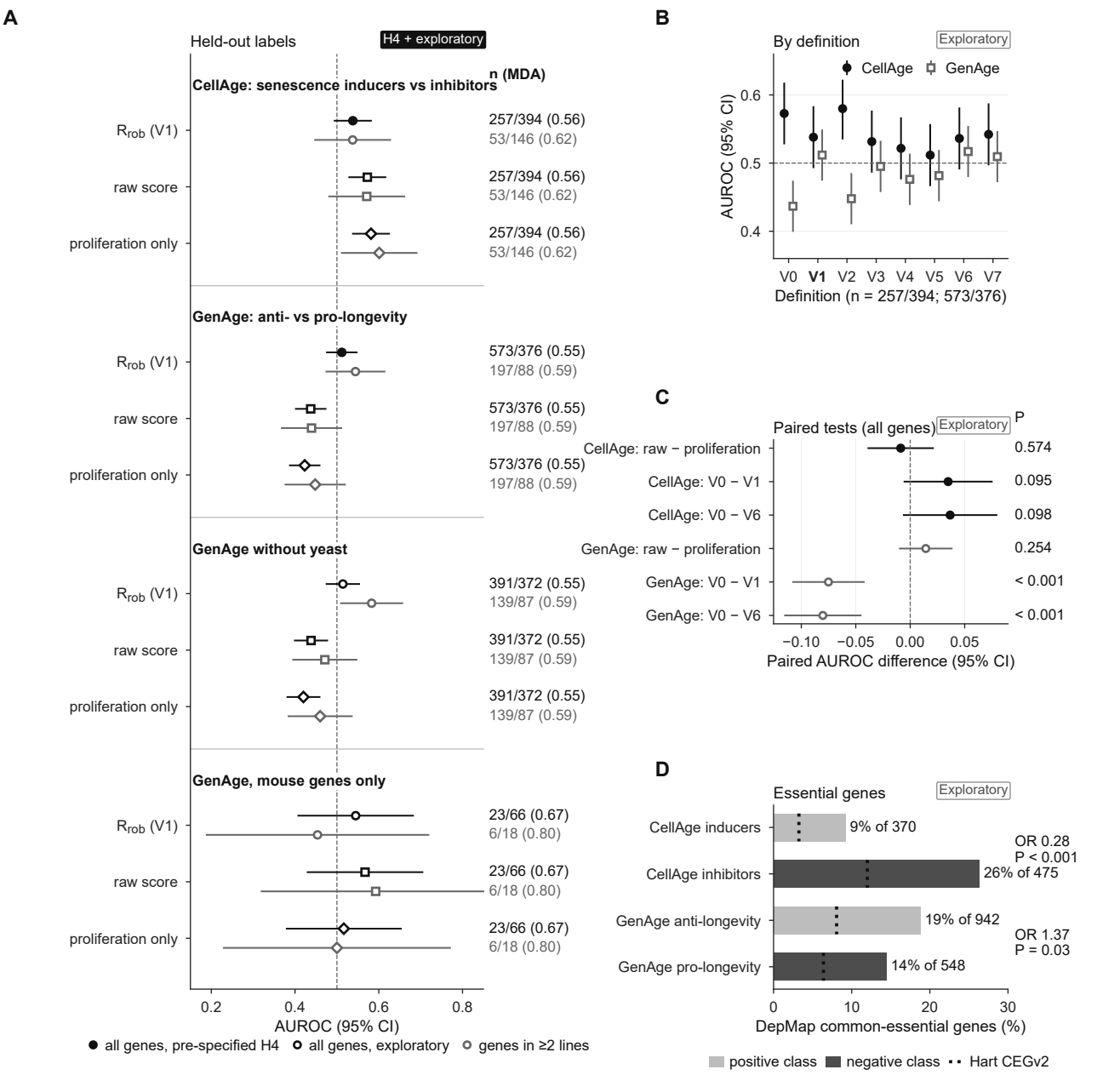

A
Held-out labels
H4 + exploratory
n (MDA)
CellAge: senescence inducers vs inhibitors
Rrob (V1)
raw score
proliferation only
257/394 (0.56)
53/146 (0.62)
257/394 (0.56)
53/146 (0.62)
257/394 (0.56)
53/146 (0.62)
GenAge: anti- vs pro-longevity
573/376 (0.55)
197/88 (0.59)
573/376 (0.55)
197/88 (0.59)
573/376 (0.55)
197/88 (0.59)
GenAge without yeast
391/372 (0.55)
139/87 (0.59)
391/372 (0.55)
139/87 (0.59)
391/372 (0.55)
139/87 (0.59)
GenAge, mouse genes only
23/66 (0.67)
6/18 (0.80)
23/66 (0.67)
6/18 (0.80)
23/66 (0.67)
6/18 (0.80)
0.2
0.4
0.6
0.8
AUROC (95% CI)
all genes, pre-specified H4
all genes, exploratory
genes in ≥2 lines
B
By definition
Exploratory
CellAge
GenAge
AUROC (95% CI)
0.6
0.5
0.4
V0
V1
V2
V3
V4
V5
V6
V7
Definition (n = 257/394; 573/376)
C
Paired tests (all genes)
Exploratory
P
CellAge: raw − proliferation
CellAge: V0 − V1
CellAge: V0 − V6
GenAge: raw − proliferation
GenAge: V0 − V1
GenAge: V0 − V6
0.574
0.095
0.098
0.254
< 0.001
< 0.001
−0.10
−0.05
0.00
0.05
Paired AUROC difference (95% CI)
D
Essential genes
Exploratory
CellAge inducers
CellAge inhibitors
GenAge anti-longevity
GenAge pro-longevity
9% of 370
26% of 475
19% of 942
14% of 548
OR 0.28
P < 0.001
OR 1.37
P = 0.03
0
10
20
30
DepMap common-essential genes (%)
positive class
negative class
Hart CEGv2